

Calculus

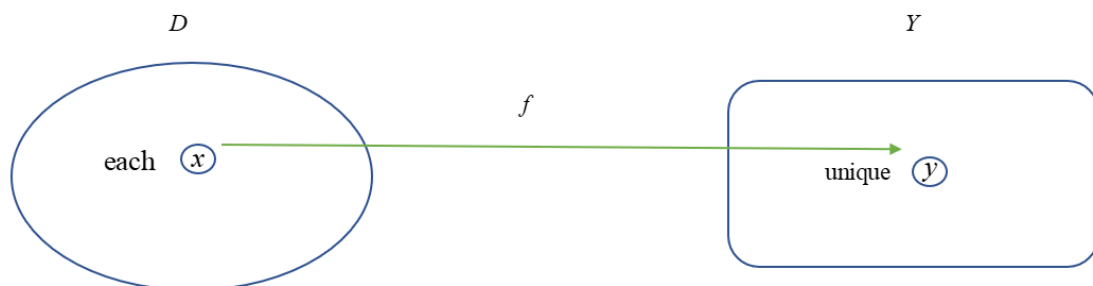
Calculus, a mathematical concept, is the study of differentiation and integration of functions in advanced mathematics. It is a fundamental discipline in mathematics, mainly including limits, differentiation, integration, and their applications. The research object is functions and the key tool for studying calculus is the limit.

The basic theory of calculus indicates that differentiation and integration are inverses of each other. This basic theory also provides a method of using algebra to solve many integral problems, that is, replacing limit operations with indefinite integral methods. This theory can also solve problems with differential equations, solving integrals of unknowns, and differential problems are ubiquitous in the field of science

Integrating has been extended to fields such as differential equations, vector analysis, variational methods, complex analysis, time-domain differentiation, and differential topology.

Functions are fundamental to the study of calculus. So we review what functions are and how they are pictured as graphs.

DEFINITION: A function f from a set D to a set Y is a **rule** that assigns a unique (single) element $y \in Y$ to each element $x \in D$. In this case, we say that “ y is a function of x ” and write this symbolically as $y = f(x)$.



The set D of all possible input values is called the **domain** of the function. The set

$$R = \{y \mid y = f(x), \forall x \in D\}$$

is called the **range** of the function.

Example 1: Let $D = \{1, 2, 3, 4\}$, $Y = \{a, b, c, d\}$. Define the rule as follows:

$$1) \quad f : \begin{cases} 1 \rightarrow a \\ 2 \rightarrow b \\ 3 \rightarrow d \\ 4 \rightarrow c \end{cases}.$$

$$2) \quad f : \begin{cases} 1 \rightarrow a \\ 2 \rightarrow b \\ 3 \rightarrow d \\ 3 \rightarrow c \end{cases}.$$

$$3) \quad f : \begin{cases} 1 \rightarrow a \\ 2 \rightarrow b, a \\ 3 \rightarrow d \\ 4 \rightarrow c \end{cases}.$$

$$4) \quad f : \begin{cases} 1 \rightarrow a \\ 2 \rightarrow b \\ 3 \rightarrow c \\ 4 \rightarrow c \end{cases}.$$

Determine if f is a function. If yes, find the range.

Example 2: The equation $A = \pi r^2$ is a rule that calculates the area A of a circle from its radius r (so r , interpreted as a length, can only be positive in this formula).

When we define a function $y = f(x)$ with a formula and the domain is not stated explicitly or restricted by context, the domain is assumed to be the largest set of real x -values for which the formula gives real y -values, which is called the **natural domain**. That is

$$\text{natural domain} = \{x \in \mathbb{R} \mid f(x) \text{ is a real number}\}.$$

Example 3: Let's verify the natural domains and associated ranges of some simple

functions.

$$1) \quad y = x^2 .$$

$$2) \quad y = \sqrt{x} .$$

$$3) \quad y = \frac{1}{x} .$$

$$4) \quad y = \sqrt{4-x} .$$

$$5) \quad y = \sqrt{1-x^2} .$$

If f is a function with domain D , its **graph** is the set of

$$\{(x, f(x)) \mid \forall x \in D\} .$$

Example 4: Graph the function $y = x + 2$ over $\mathbb{R}$.

Example 5: Graph the function $y = x^2$ over the interval $[-2, 2]$.

Ch 2 Limits and Continuity

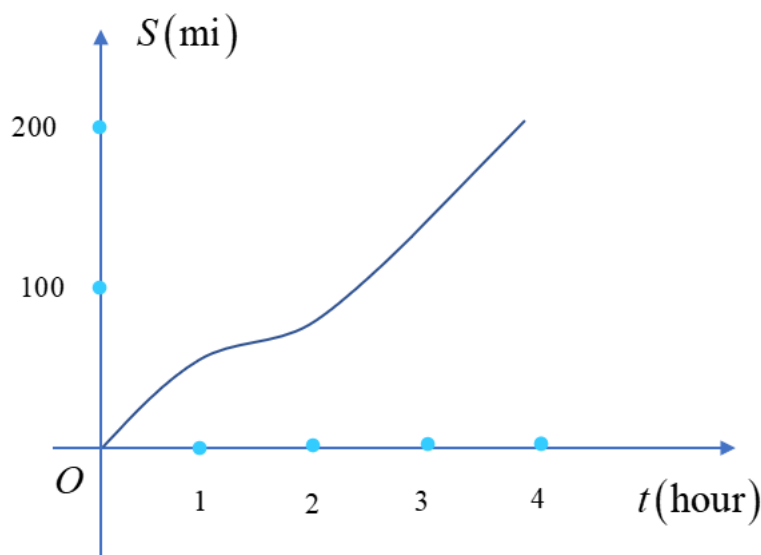
Mathematicians of the seventeenth century were keenly interested in the study of motion for objects on or near the earth and the motion of planets and stars. This study involved both the speed of the object and its direction of motion at any instant, and they knew the direction at a given instant was along a line tangent to the path of motion. The concept of a limit is fundamental to finding the velocity of a moving object and the tangent to a curve. In this chapter we develop the limit, first intuitively and then formally. We use limits to describe the way a function varies.

2.1 Rates of Change and Tangents to Curves

Suppose you take a car trip and record the distance that you travel every few minutes. The distance S you have traveled is a function of the time t :

$$S(t) = \text{total distance traveled at time } t.$$

Now, let the function S be the following



The graph shows that that $t = 1$ hour, $S = 50$ mi; $t = 2$ hour, $S = 75$ mi; $t = 3$ hour, $S = 140$ mi; $t = 4$ hour, $S = 200$ mi. To find your average speed between any two points on the trip, we divide the distance traveled by the time elapsed.

For example, the average speed between 1 hour and 4 hour is

$$\text{average speed} = \frac{\text{distance traveled}}{\text{time elapsed}} = \frac{150}{3} = 50 \text{ mi/h}.$$

If we use function notation, we get

$$\text{average speed} = \frac{S(4) - S(1)}{4 - 1} = \frac{200 - 50}{3} = 50 \text{ mi/h}.$$

Example 1 Let the distance(feet) of a falling rock satisfy the function

$$y = 16t^2.$$

What is its average speed

(a) during the first 2 sec of fall?

(b) during the 1-sec interval between second 1 and second 2?

Solution

$$(a) \text{ average speed} = \frac{\text{distance fallen}}{\text{time elapsed}} = \frac{y(2) - y(0)}{2 - 0} = \frac{16(2)^2 - 0}{2} = 32 \text{ ft/sec}.$$

$$(b) \text{ average speed} = \frac{\text{distance fallen}}{\text{time elapsed}} = \frac{y(2) - y(1)}{2 - 1} = \frac{16(2)^2 - 16(1)^2}{1} = 48 \text{ ft/sec}.$$

Note We can calculate the average speed of the rock over a time interval $[t_0, t_0 + h]$, having length $\Delta t = (t_0 + h) - (t_0) = h$, as

$$\frac{\Delta y}{\Delta t} = \frac{16(t_0 + h)^2 - 16t_0^2}{h} \text{ ft/sec}.$$

We cannot use the average formula to calculate the “instantaneous” speed at the exact moment t_0 by simply substituting $h = 0$, because we cannot divide by zero. But we *can* use it to calculate average speeds over shorter and shorter time intervals starting at t_0 . For example, taking $h = 0.01, 0.001, 0.001\cdots$ and $t_0 = 1$, we get the speed 32.16, 32.016, 32.0016 $\cdots$. It seems to approach a limiting value of 32 as the length of the interval decreases. This suggests that the rock is falling at a speed of 32 ft/sec at $t_0 = 1$ sec.

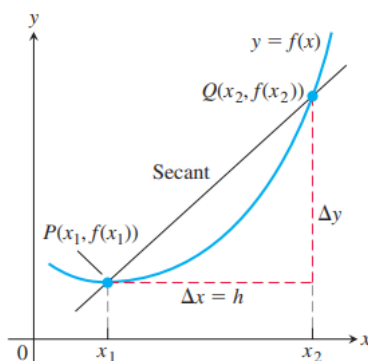
The average speed of an object is an example of a more general idea which we discuss next.

① The average rate of change

In College Algebra and Trigonometry, we learned that the average rate of change of the function $y = f(x)$ over $[x_1, x_2]$ is

$$\frac{\text{change in } y}{\text{change in } x} = \frac{\Delta y}{\Delta x} = \frac{f(x_2) - f(x_1)}{x_2 - x_1}.$$

Geometrically, the rate of change of f over $[x_1, x_2]$ is the slope of the line through the points $P(x_1, f(x_1))$ and $Q(x_2, f(x_2))$. In geometry, a line joining two points of a curve is a secant to the curve. Thus, the average rate of change of f from x_1 to x_2 is identical with the slope of secant PQ .



Let's consider what happens as the point Q approaches the point P along the curve, so the length h of the interval over which the change occurs approaches zero. We will see that this procedure leads to defining the slope of a curve at a point.

② Defining the Slope of a Curve

We know what is meant by the slope of a straight line. But what is meant by the slope of a curve at a point P on the curve?

If there is a tangent line to the curve at P , it would be reasonable to identify the slope of the tangent as **the slope of the curve at P** .

So we need a precise meaning for the tangent at a point on a curve.

For circles, tangency is straightforward. A line L is tangent to a circle at a point P if L passes through P perpendicular to the radius at P . Such a line just touches the circle.

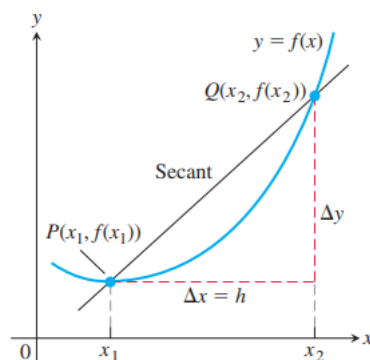
But what does it mean to say that a line L is tangent to some other curve C at a point P ?

To define tangency for general curves, we need an approach that takes into account the behavior of the secants through P and nearby points Q as Q moves toward P along the curve. Here is the idea.

(1) Start with the slope of the secant PQ .

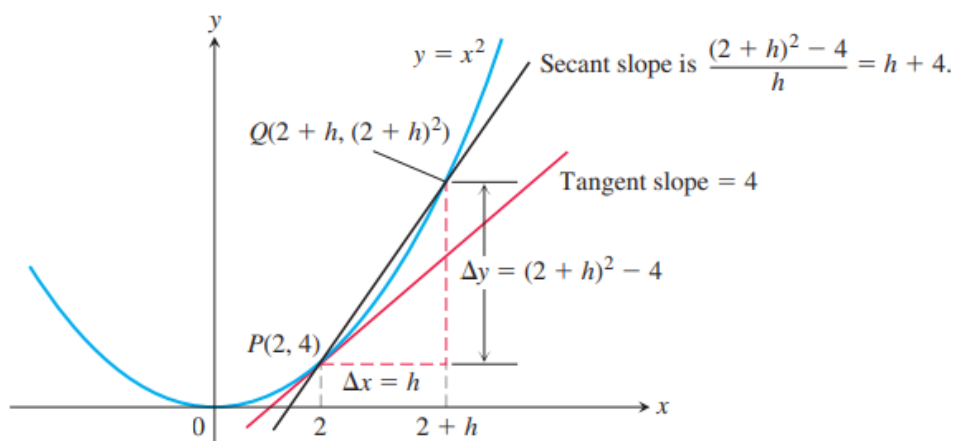
(2) Investigate the limiting value of the secant slope as Q approaches P along the curve.

(3) If the limit exists, take it to be the slope of the curve at P and define the tangent to the curve at P to be the line through P with this slope.



Example 2 Find the slope of the parabola $y = x^2$ at the point $P(2, 4)$.

Solution We begin with a secant line through $P(2, 4)$ and $Q(2 + h, (2 + h)^2)$ nearby.



We then write an expression for the slope of the secant PQ and investigate what happens to the slope as Q approaches P along the curve:

$$\begin{aligned}\text{Secant slope} &= \frac{\Delta y}{\Delta x} = \frac{(2+h)^2 - 2^2}{h} = \frac{h^2 + 4h + 4 - 4}{h} \\ &= \frac{h^2 + 4h}{h} = h + 4.\end{aligned}$$

If $h > 0$, then Q lies above and to the right of P , as in Figure.

If $h < 0$, then Q lies to the left of P (not shown).

In either case, as Q approaches P along the curve, h approaches zero and the secant slope $h + 4$ approaches 4. We take 4 to be the parabola's slope at P .